

Diseases of the biliary system and pancreas

Fordøjelsessystem, patofysiologi og diagnostik

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DENMARK

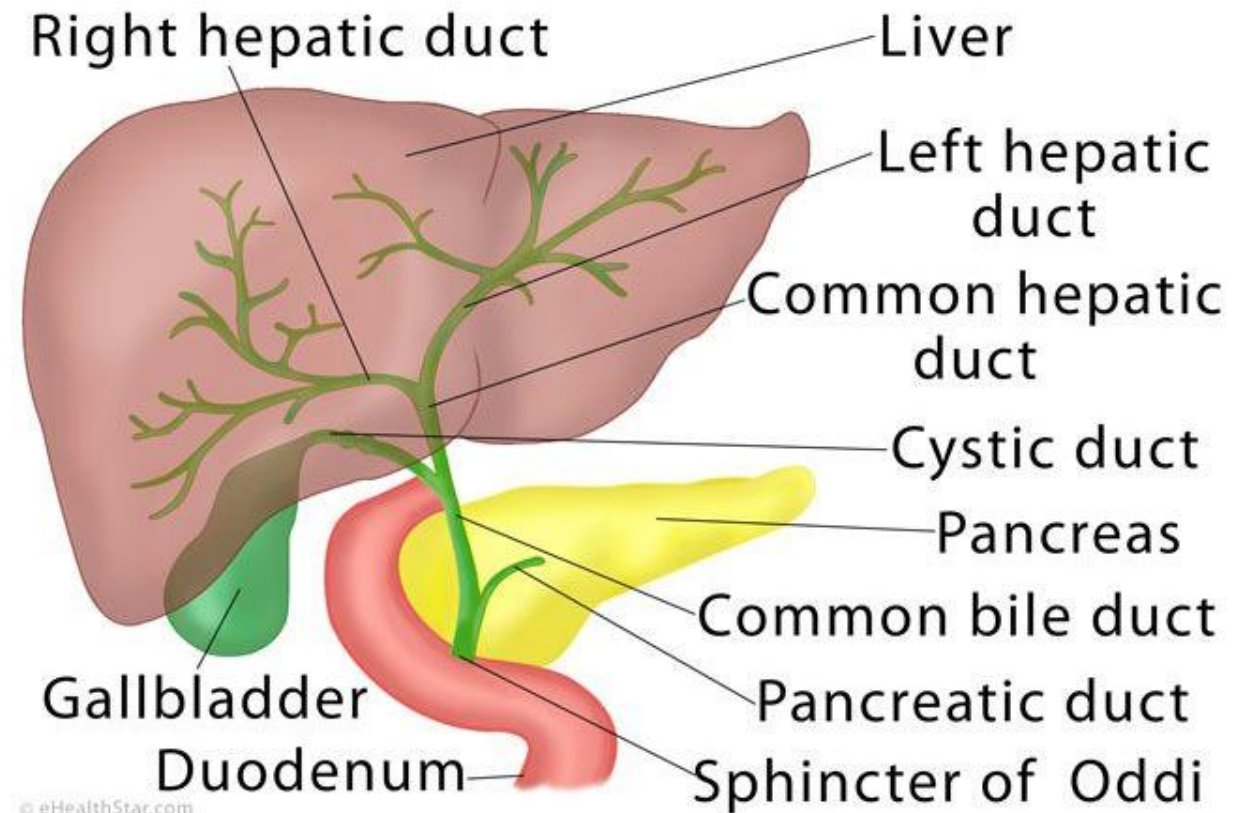


Læringsmål

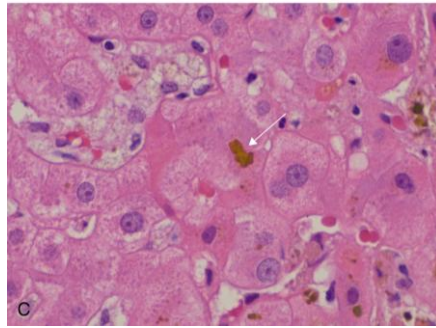
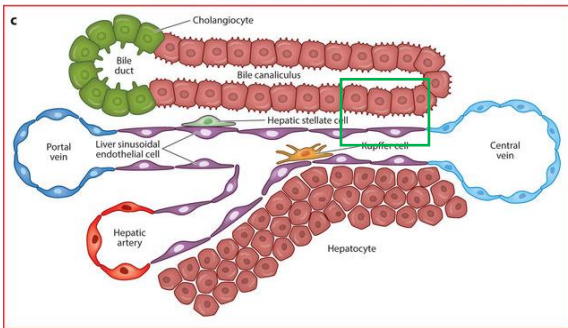
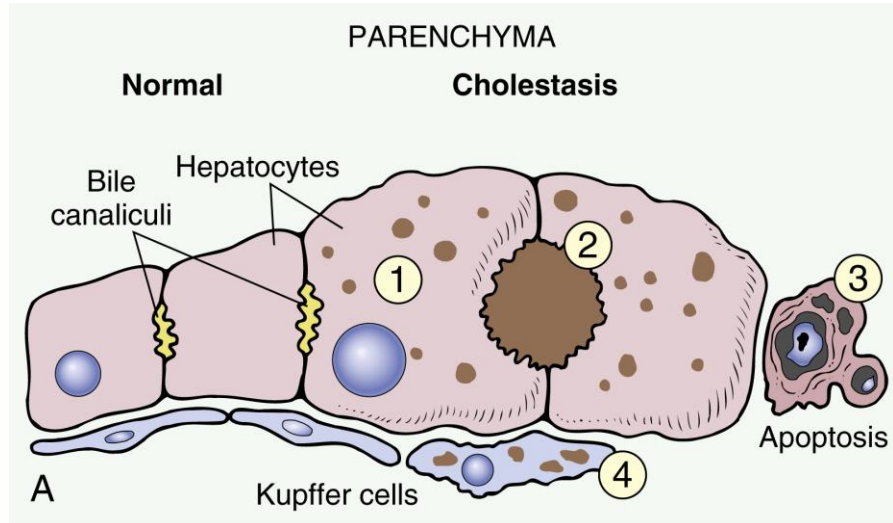
- *Beskriv de patologiske processer og konsekvenser af galdesten*
- *Gør rede for patologi af akut og kronisk betændelse i bugspytkirtlen (pankreatit)*
- *Beskriv udviklingen og konsekvenser af kræft i bugspytkirtlen og galdesystemet*

Biliary system

- Gallbladder and biliary tree
- Gallbladder: storage, concentration and release of bile
- Bile contains: cholesterol, phospholipids, bile salts, conjugated bilirubin, electrolytes
- Discharge bile in the duodenum to contribute to fat absorption



Cholestasis

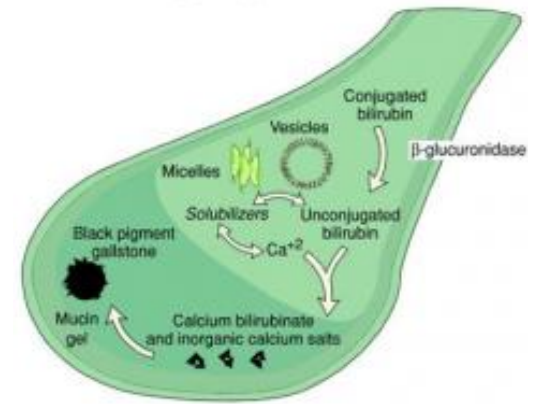
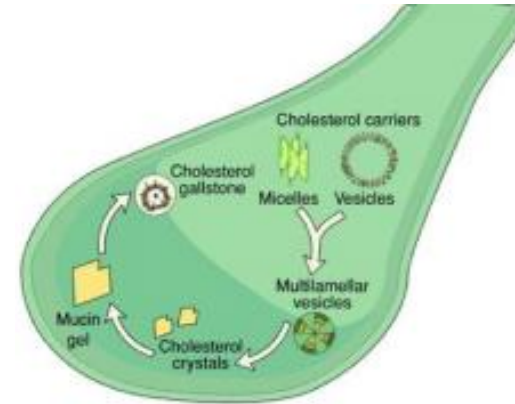


- Accumulation of bile in the liver due to obstruction (intrahepatic or extrahepatic bile ducts)
- Increases conjugated bilirubin in serum (*direct bilirubin*)
- Cholestatic hepatocytes (1) are enlarged with dilated canaliculi spaces (2) . Apoptotic cells (3) may be seen, and Kupffer cells (4) frequently contain bile pigments.

Cholelithiasis

Formation of stones in the gallbladder

- Pathogenesis
 - Cholesterol stones
 - Supersaturation of cholesterol
 - Hypomotility of gallbladder
 - Increased mucin secretion
 - Pigment stones
 - Precipitation of calcium bilirubinate
 - Black pigment stones: increased unconjugated bilirubin in bile (e.g. chronic hemolysis)
 - Brown pigment stones: biliary infection causing bilirubin deconjugation
- Symptoms
 - 70-80%: asymptomatic
 - 20-30%: inflammation of biliary tree/pancreatitis/ileus



Cholesterol stones

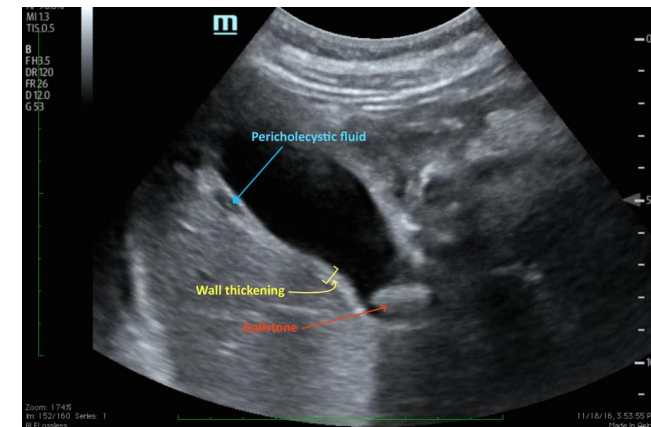
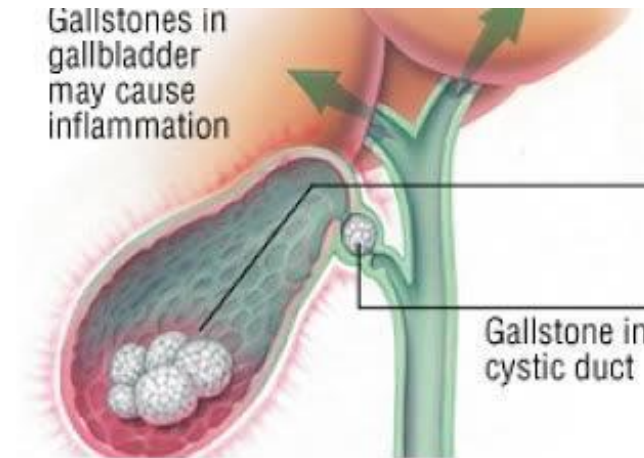


Pigment stones

Cholecystitis

Acute

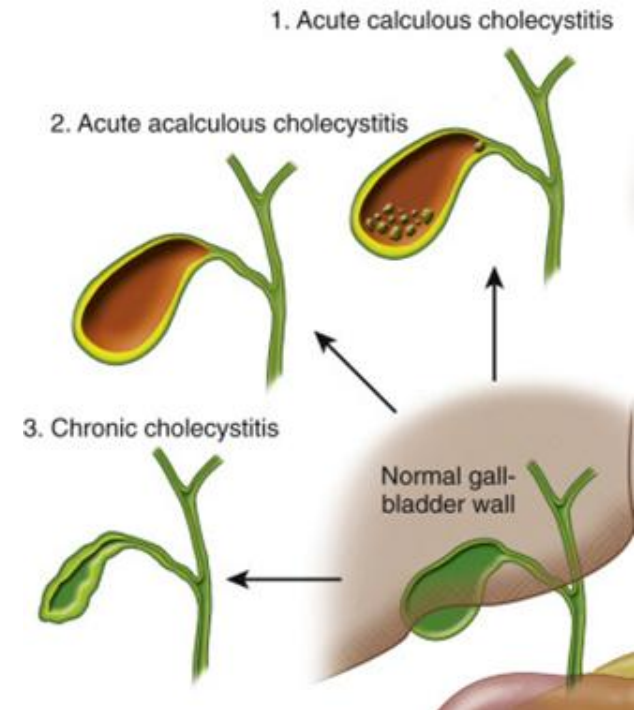
- Etiology: Obstruction of the neck of the gallbladder or the cystic duct by gallstones (90% of cases)
- Clinical features
 - Severe upper abdominal pain
 - Nausea
 - Fever
 - Leukocytosis
 - (+) Murphy's sign



Cholecystitis

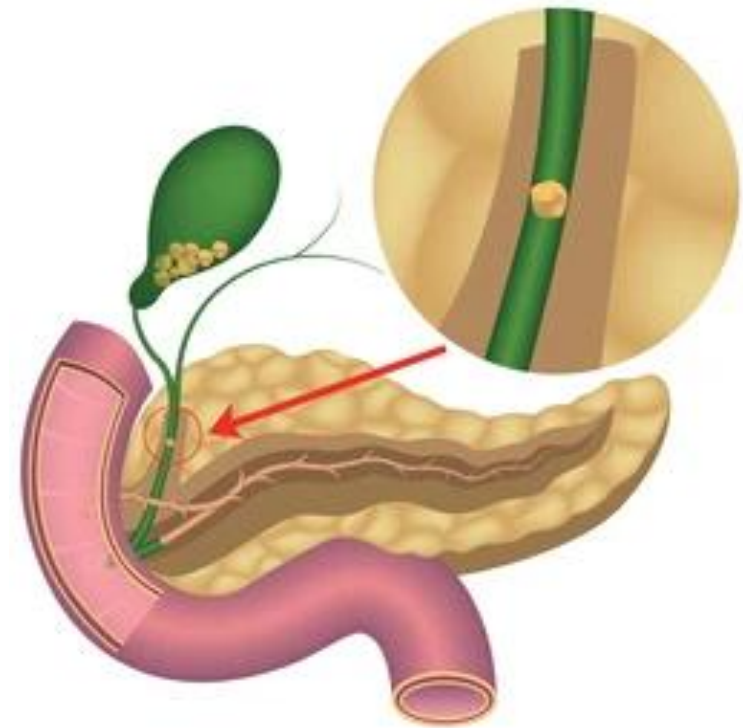
Chronic

- Etiology: chronic irritation of the mucosa by supersaturated bile/stones
- Clinical features:
 - Asymptomatic for years
 - Recurrent attacks of abdominal pain
 - Nausea
 - Vomiting
 - Intolerance of fatty foods
- Long-term risk: carcinoma of the gallbladder



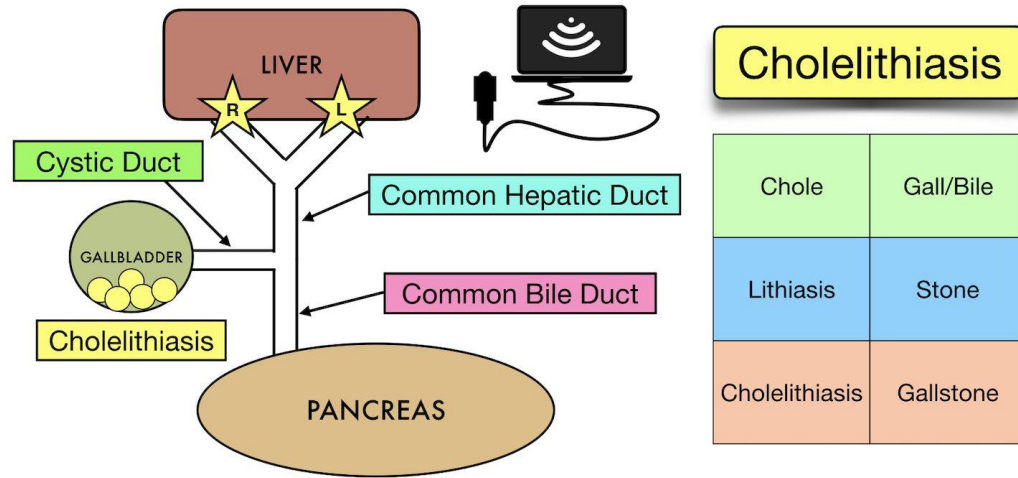
Disorders of the extrahepatic bile ducts

- Biliary atresia: congenital condition where bile ducts do not develop normally
- Choledocholithiasis: Presence of stones within the biliary tree
- Cholangitis: Inflammation of the bile ducts caused by infection or immune reactions
 - *Ascending cholangitis*: infection of the biliary tract by enteric bacteria
 - Autoimmune cholangiopathies: primary biliary cholangitis and primary sclerosing cholangitis



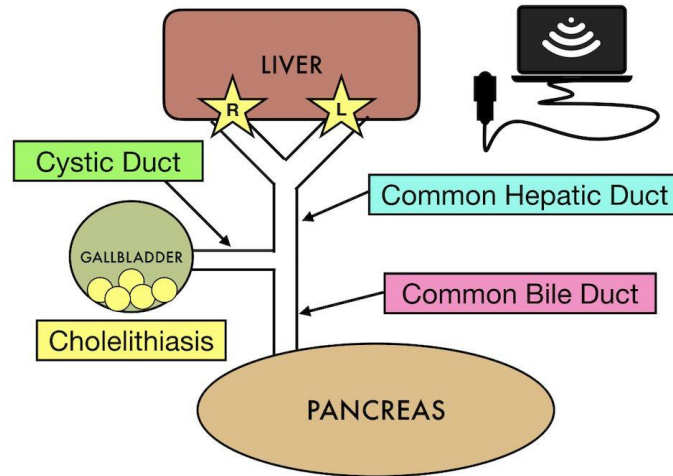
Choledocholithiasis

Summary of gallstone-related diseases in the biliary system



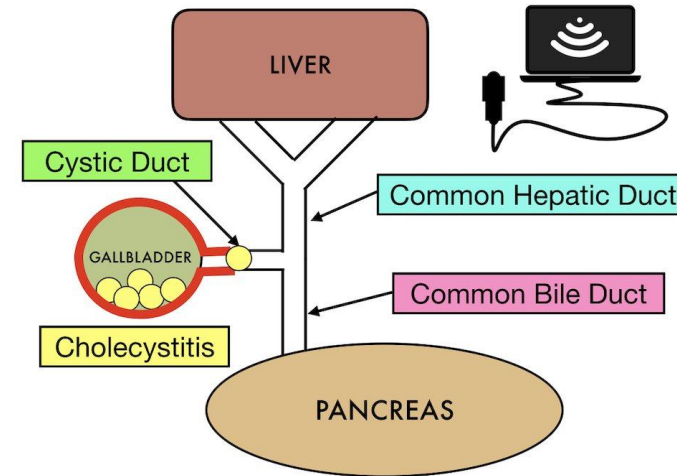
- Cholelithiasis (Gallstones) = Hardened deposits in the biliary tract, usually in the gallbladder
- Gallstones don't necessarily cause symptoms, but problems can occur when they move or obstruct

Summary of gallstone-related diseases in the biliary system



Cholelithiasis	
Chole	Gall/Bile
Lithiasis	Stone
Cholelithiasis	Gallstone

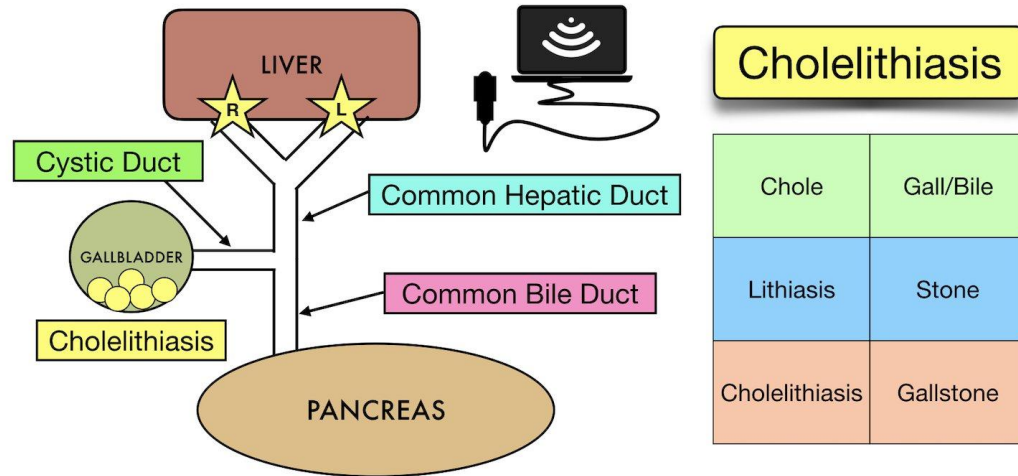
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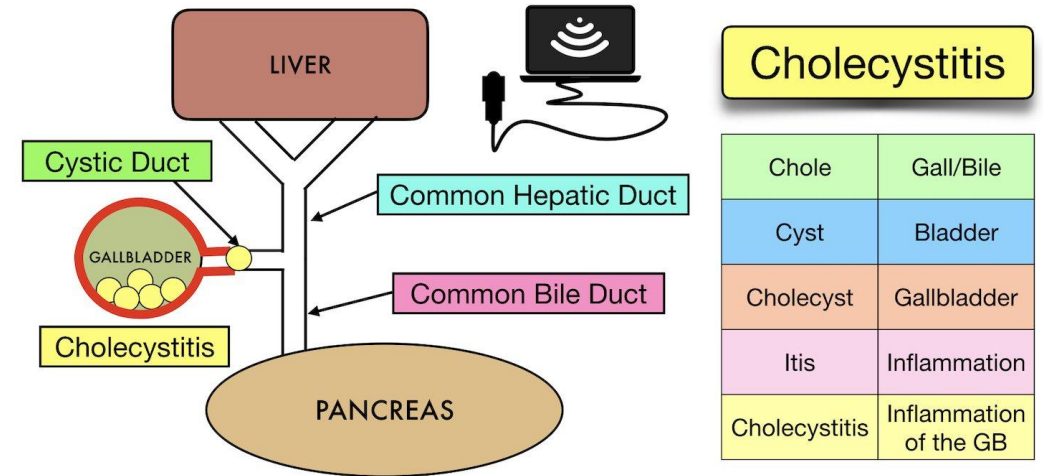
Cholecystitis	
Chole	Gall/Bile
Cyst	Bladder
Cholecyst	Gallbladder
Itis	Inflammation
Cholecystitis	Inflammation of the GB

- Biliary Colic = Abdominal pain caused by gallstones obstructing/traveling through biliary ducts
- Cholecystitis = Inflammation of the gallbladder

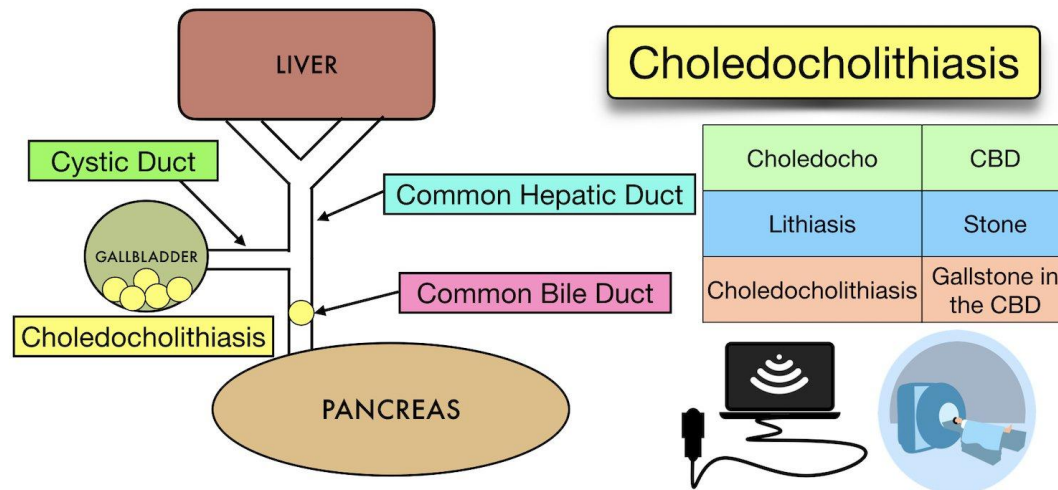
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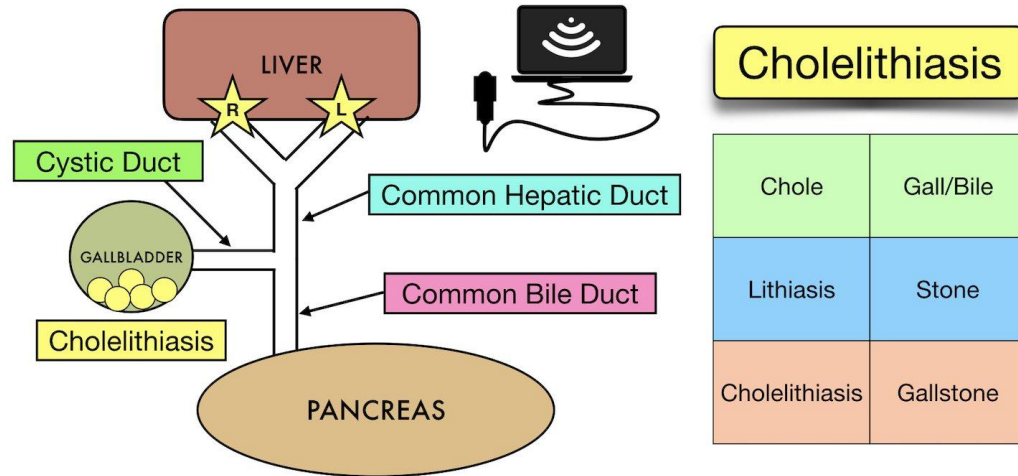


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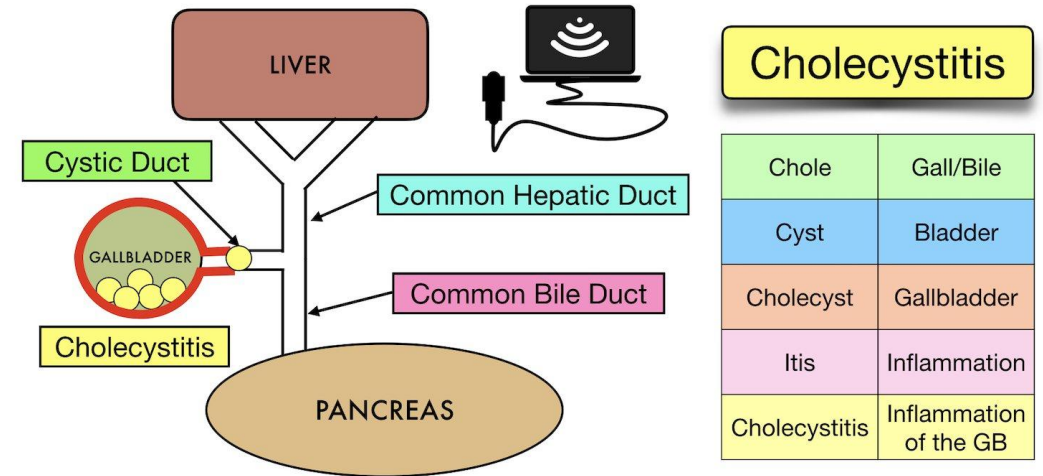


- **Choledocholithiasis** = The presence of gallstone(s) in the common bile duct (CBD)
- May experience abdominal pain (RUQ), jaundice, nausea and vomiting

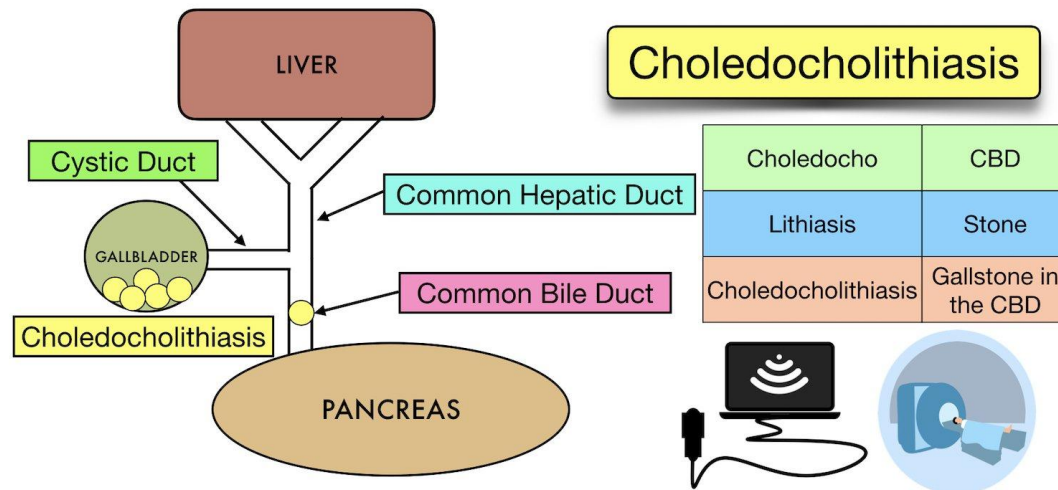
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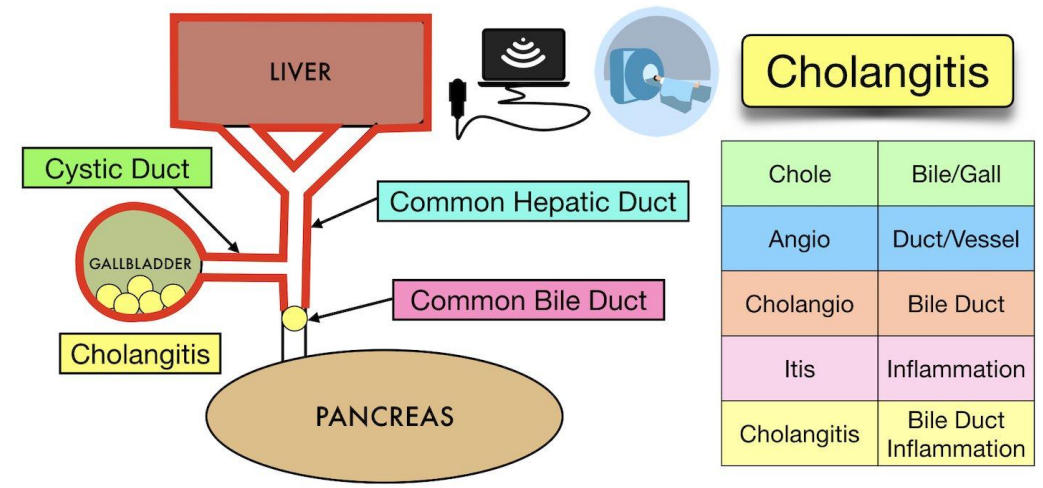
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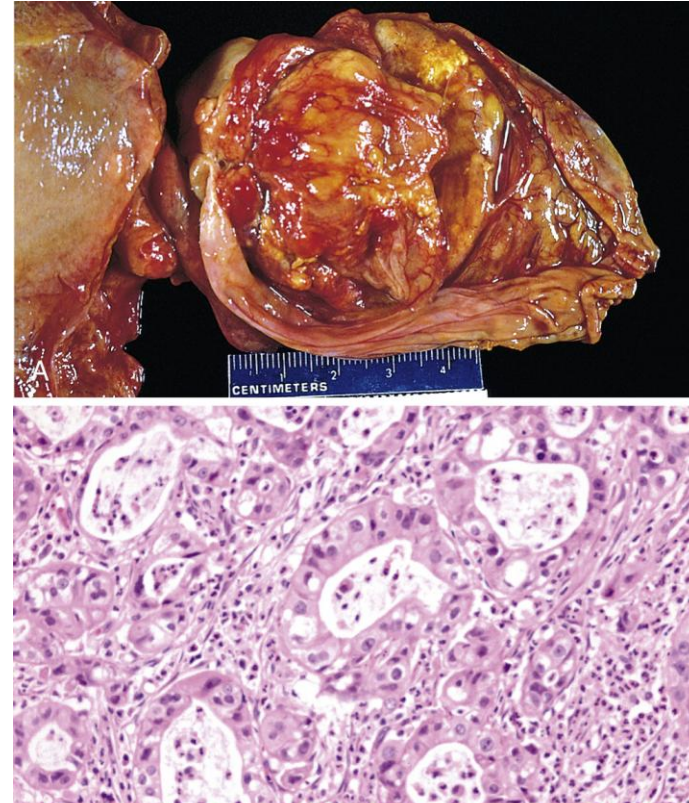
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- May experience abdominal pain (RUQ), jaundice, nausea and vomiting



- **Cholangitis** = Inflammation of the bile ducts, usually caused by an obstructing gallstone in the CBD
- May cause fever, abdominal pain, jaundice, altered mental status, nausea and vomiting

Gallbladder adenocarcinomas

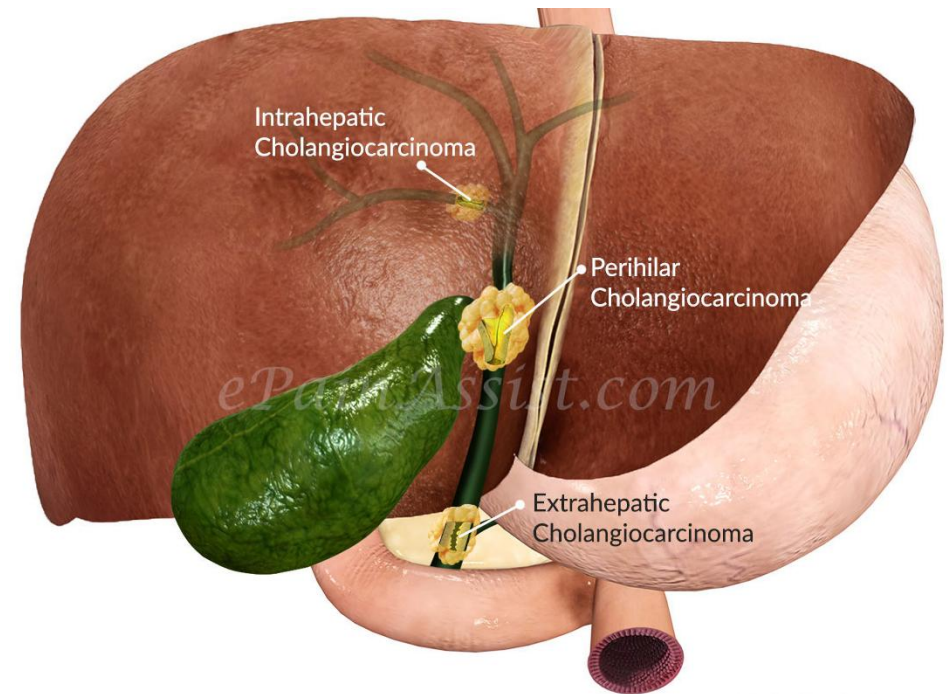
- Adenocarcinomas (transformation of **cholangiocytes**)
- The most common malignancy of the extrahepatic biliary tract
- Usually asymptomatic
- Risk factor: **chronic cholelithiasis and chronic cholecystitis**
- Poor prognosis



Malignant glands are seen infiltrating a densely fibrotic gallbladder wall

Cholangiocarcinomas

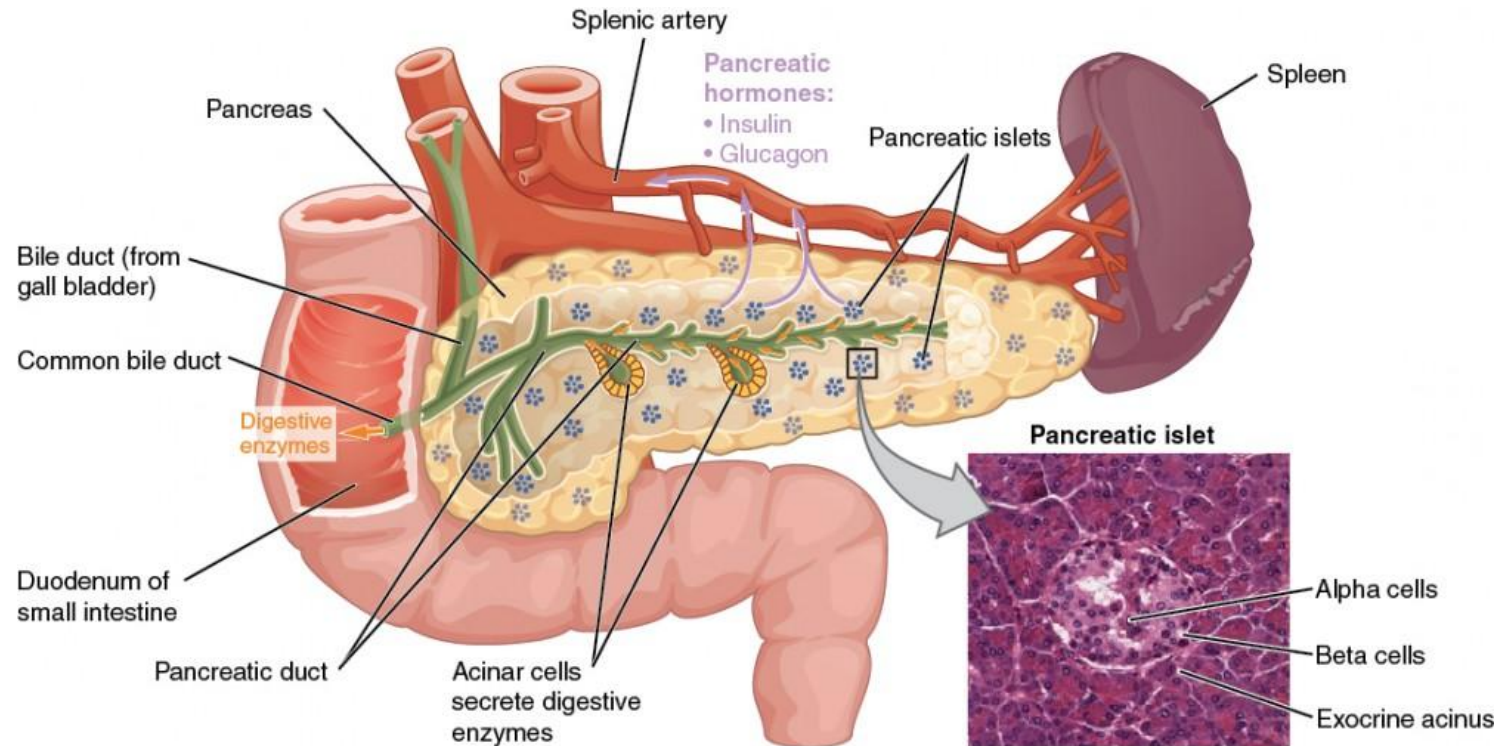
- Adenocarcinomas (transformation of **cholangiocytes**)
- Intra- or extrahepatic
- Partial or complete obstruction of the bile ducts
- Risk factor: primary sclerosing cholangitis
- Poor prognosis



Biliary tract pathology: Key takeaways

- Cholestasis: Impaired bile flow due to obstruction (intra- or extrahepatic).
- Cholelithiasis (Gallstones)
- Cholecystitis:
 - Acute: Caused by gallstone obstruction (90%); presents with pain, fever, nausea, Murphy's sign.
 - Chronic: Due to long-term irritation; may be asymptomatic or cause recurrent pain and food intolerance.
- Extrahepatic disorders
 - Biliary atresia
 - Choledocholithiasis: Stones in the bile ducts.
 - Cholangitis: Inflammation due to infection or autoimmune causes.
- Malignancies
 - Gallbladder adenocarcinoma: Originates from cholangiocytes lining the mucosa.
 - Cholangiocarcinoma: Adenocarcinoma of intra- or extrahepatic bile ducts.

Pancreas



Exocrine pancreas:

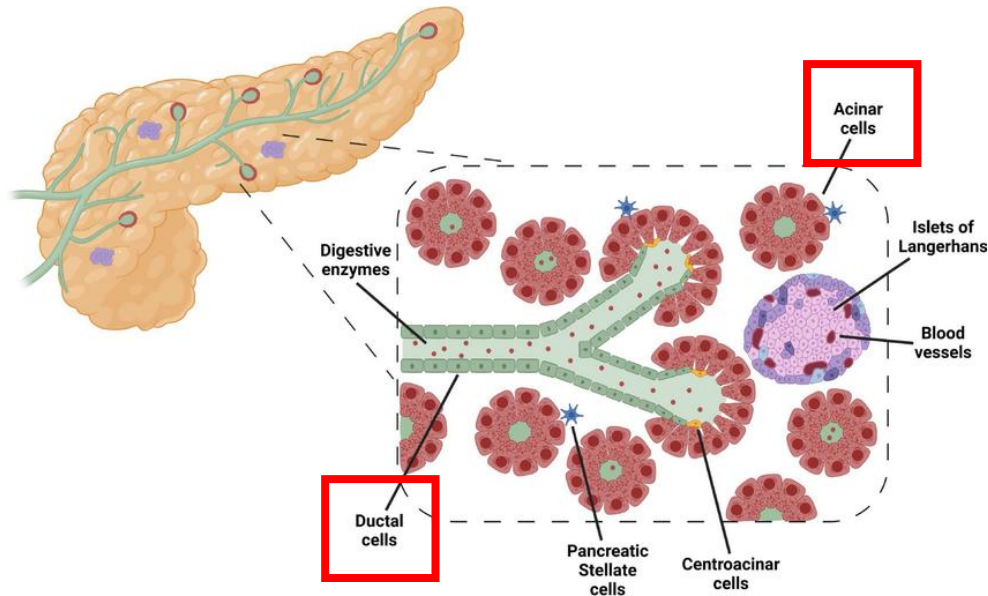
- Proteases (trypsin, chymotrypsin, carboxylpeptidase, in inactive form)
- Amylase, lipase, cholesterol esterase, phospholipase
- Trypsin inhibitor
- Water and bicarbonate

Pancreatitis (acute)

- Inflammation of the pancreas parenchyma
- Etiology: Alcoholism and gallstone disease represent 80% of the causes
- Clinical features:
 - Nausea/vomiting
 - Abdominal pain radiating to the back
 - Abdominal tenderness
 - Elevated plasma levels of **amylase** and lipase

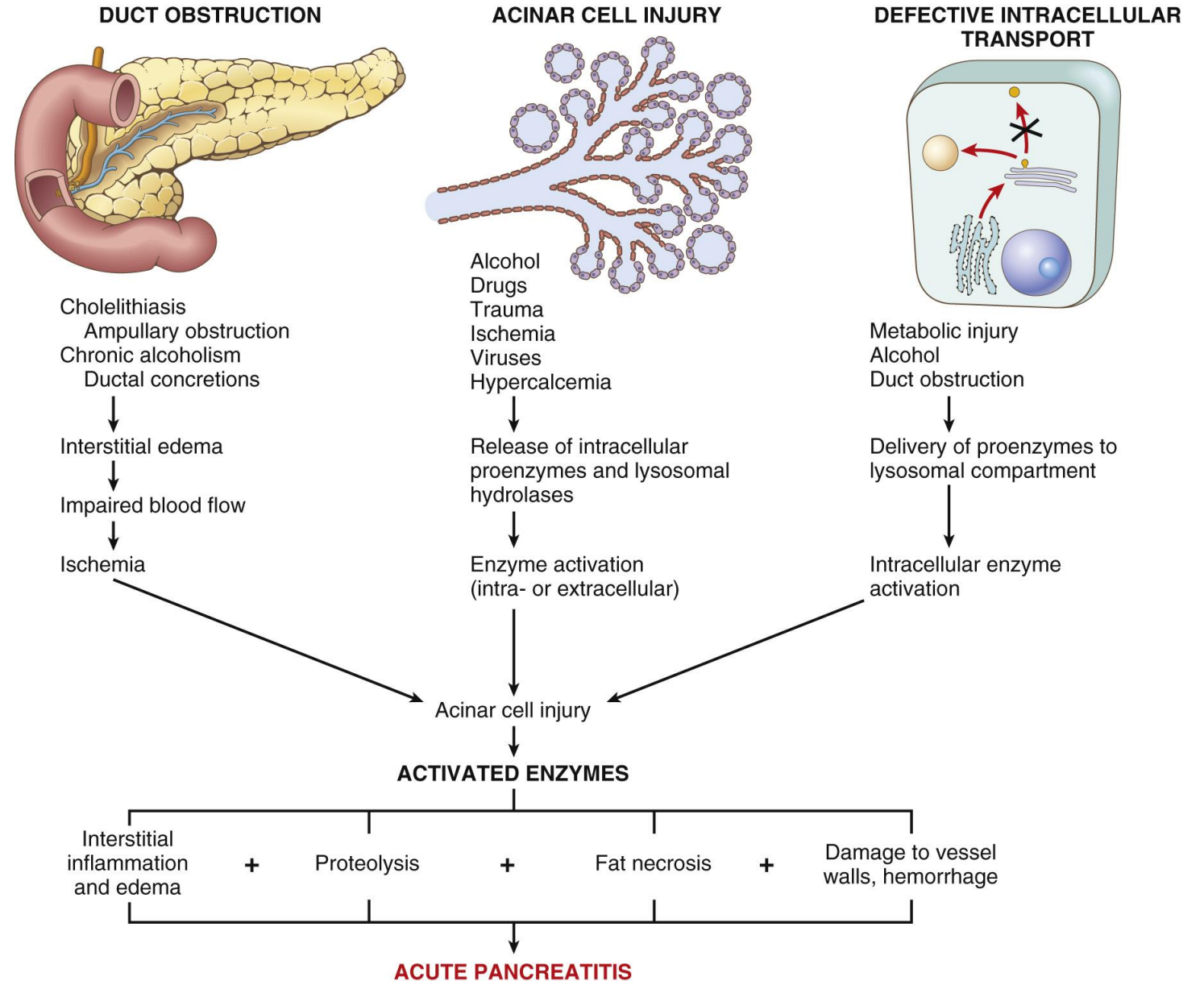


Patterns of pancreatic tissue injury



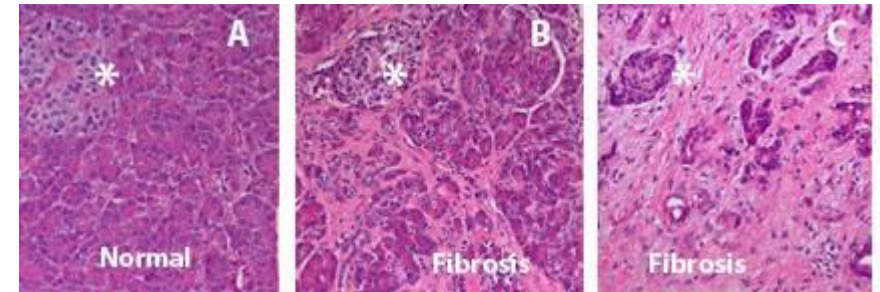
- **Acinar cell injury:** Central in acute pancreatitis. Premature activation of digestive enzymes leads to autodigestion, fat necrosis, and cell death.
- **Ductal cell injury:** In chronic pancreatitis, ductal obstruction and dysfunction promote fibrosis and ductal necrosis.
- **Neoplastic transformation:** Pancreatic ductal adenocarcinoma (PDAC) often arises from transformed ductal cells, especially in the context of **chronic inflammation** and acinar-to-ductal metaplasia (ADM),

Acute pancreatitis: Proposed pathogenesis

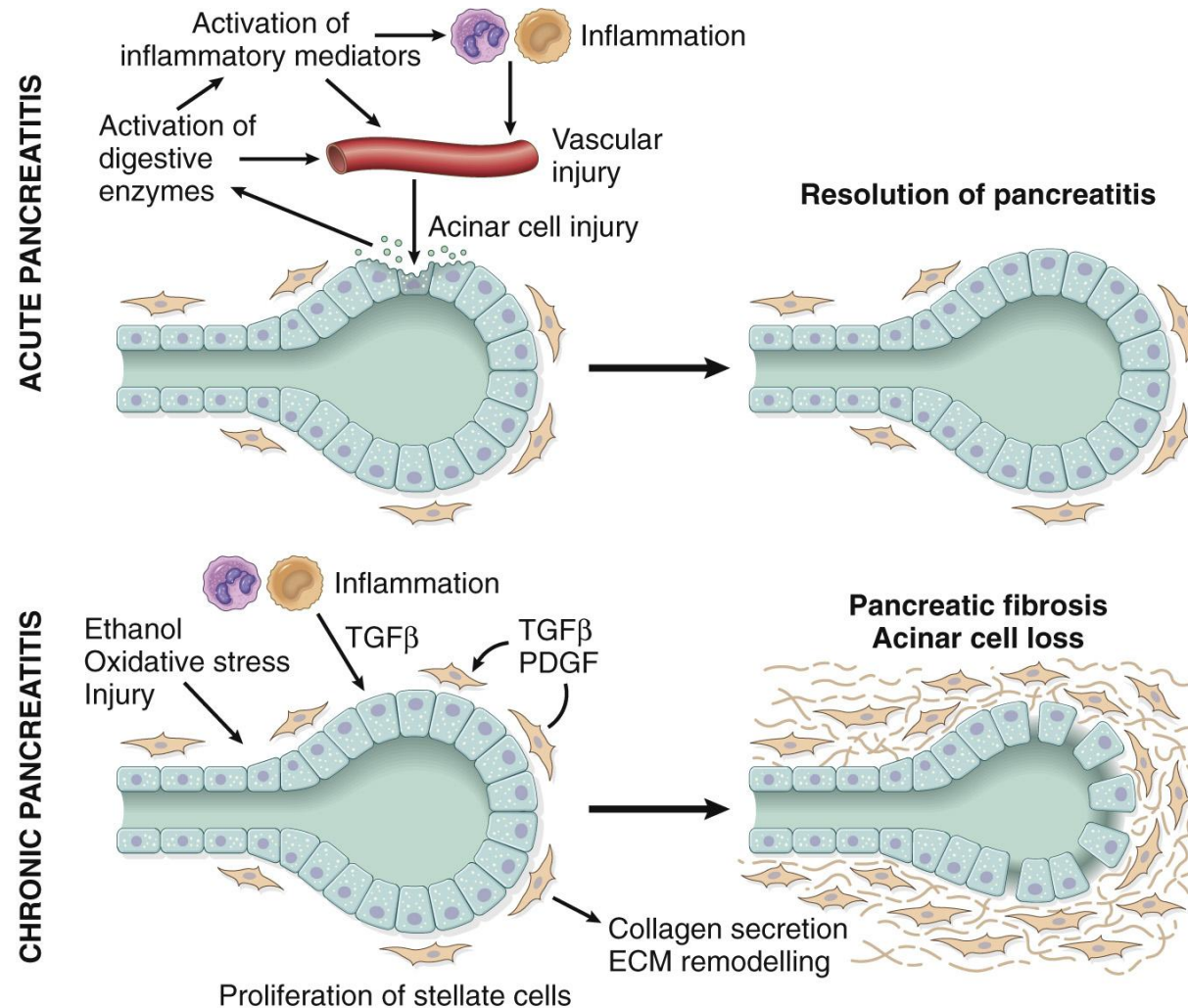


Chronic pancreatitis

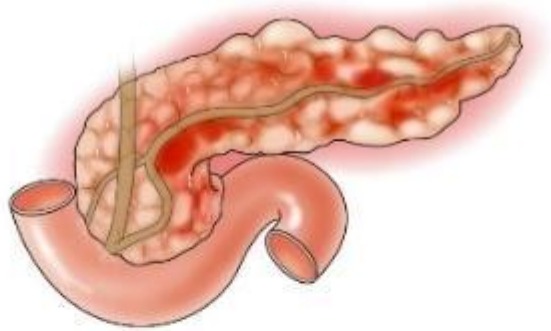
- Repeated bouts of acute pancreatitis
- Chronic inflammation with irreversible changes
- Accompanied of reduced pancreas function
- Smoking is one of the major risk factors for progression into chronic disease



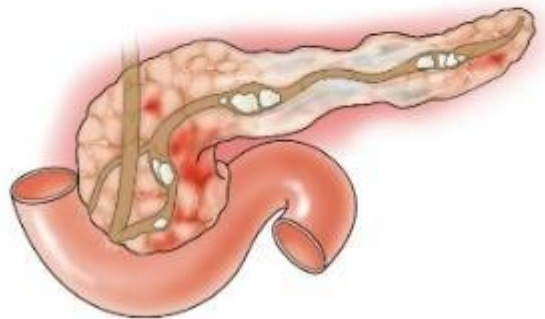
Histologic features of acute vs. chronic pancreatitis



Features of acute vs. chronic pancreatitis



Acute pancreatitis

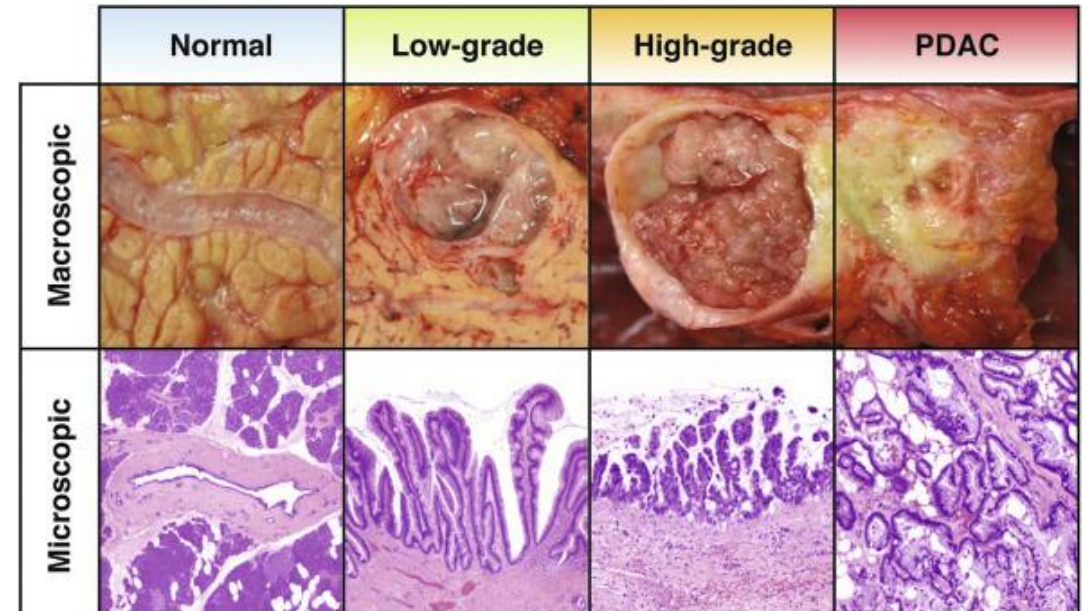
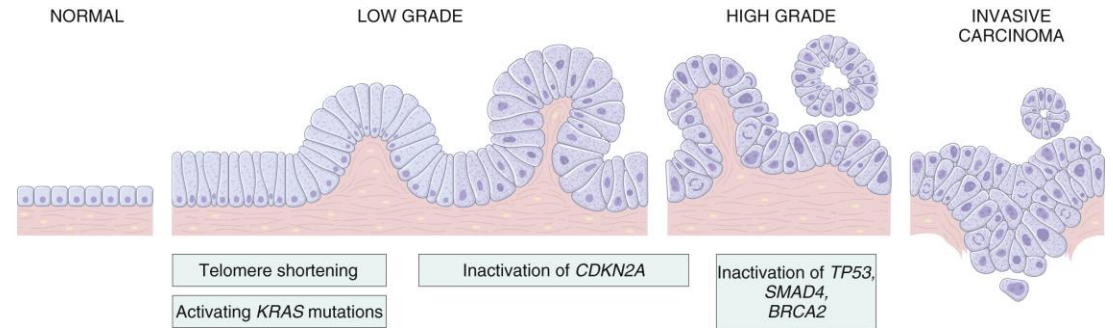


Chronic pancreatitis

Feature	Acute Pancreatitis	Chronic Pancreatitis
Pancreatic size	Enlarged	Atrophic
Calcifications	Rare	Common
Ductal changes	Usually normal	Dilated, irregular
Fluid collections	Common	May have pseudocysts
Necrosis	Possible	Uncommon
Fibrosis	Absent	Present

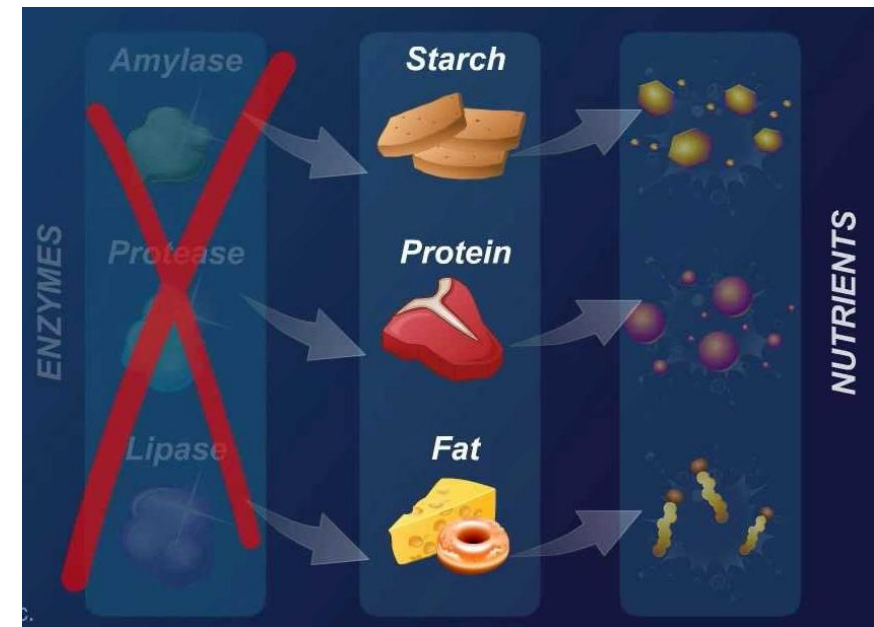
Pancreatic neoplasms

- Infiltrating pancreatic ductal adenocarcinoma (PDAC) (90%)
- From noninvasive precursor lesions (pancreatic intraepithelial neoplasia)
- Due to mutations and epigenetic alterations (oncogenes and tumor suppressor genes)
- Poor prognosis: 5-year survival rate is 10%
- Major risk: chronic pancreatitis



Primary exocrine pancreatic insufficiency

- When less than 10% of pancreatic digestive enzymes is available (lipase in particular)
- Clinical manifestation
 - Steatorrhea
 - Weight loss
- Etiology
 - Pancreatitis (mostly in its chronic form)
 - Malignancies
 - Cystic fibrosis
- Diagnosis
 - Fecal elastase-1 test (FE-1)
 - Fecal fat analysis



Pancreas disease: Key takeaways

- **Pancreatitis:**
 - Inflammation of the pancreas parenchyma.
 - Common causes: Alcoholism and gallstone disease (80%).
- **Chronic pancreatitis:**
 - Results from repeated acute pancreatitis bouts.
 - Features chronic inflammation with irreversible changes.
- **Acute vs. Chronic pancreatitis:**
 - Acute: Sudden inflammation; often reversible.
 - Chronic: Persistent inflammation; irreversible damage.
- **Pancreatic neoplasms:**
 - PDAC is the most prevalent form of pancreatic malignancy.
 - Poor prognosis
 - Chronic pancreatitis is a major risk factor for PDAC.

Literature

- Robbins Basic Pathology, af Vinay Kumar, m.fl., 10th ed. Elsevier Saunders
- Patologi, af Niels Marcussen, Flemming B. Sørensen, Susanne Holck, Torben Steiniche og Eric Santoni-Rugiu. FADL's forlag. 2. udgave (2015).
- Or any other pathology/pathophysiology books from previous semesters



<https://gutscharity.org.uk/awareness/kranky-panky/>